Personal Meeting

November 2, 2026

1 HRSID Pixel Scale

The echo simulation uses each HRSID pixel value directly as a scatterer amplitude, which requires two properties of the pixel value to be confirmed: (1) linear, not logarithmic; (2) amplitude, not amplitude squared.

Product specifications.

- HRSID is cropped from Sentinel-1B, TerraSAR-X and TanDEM-X panoramas [1].
- Sentinel-1 Level-1 GRD: $\beta^0 = DN^2/A_\beta^2$, so the stored DN is amplitude [2].
- TerraSAR-X/TanDEM-X detected products: $\beta^0 = k_s\, DN^2$, so DN is amplitude [3].
- How HRSID converted the products to 8-bit images is not documented.

Data. 975,864 homogeneous 32×32 sea patches from 1593 offshore scenes, all farther than 10 pixels from every ship polygon.

Test 1: linear vs. log. Let x be the grey value of one pixel in a homogeneous sea patch. The complex return of a pixel is a sum of many scatterers with random phases, a zero-mean complex Gaussian $s \sim \mathcal{CN}(0, \sigma^2)$, so $s = \sigma u$ with $u \sim \mathcal{CN}(0, 1)$: the backscatter only sets the scale σ. Taking the magnitude keeps the scale out, $|\sigma u| = \sigma|u|$ and $|\sigma u|^2 = \sigma^2|u|^2$, so in linear units

$$x = c\, n, \qquad \begin{cases} \text{amplitude: } c = \sigma, \quad n = |u| \sim \text{Rayleigh (Nakagami for } L \text{ looks)}, \\ \text{intensity: } c = \sigma^2, \ n = |u|^2 \sim \text{exponential (Gamma for } L \text{ looks)}, \end{cases}$$

where the distribution of n does not depend on c. Within a patch c is a constant, so the mean and the standard deviation scale with c:

$$\operatorname{mean}(x) = E[c\, n] = c\, E[n],$$
$$\operatorname{Var}(x) = E\big[(c\, n - c\, E[n])^2\big] = c^2 \operatorname{Var}(n) \ \Rightarrow\ \operatorname{std}(x) = c \operatorname{std}(n) \quad (c > 0).$$

In both cases c cancels in the ratio

$$\frac{\operatorname{std}(x)}{\operatorname{mean}(x)} = \frac{c \operatorname{std}(n)}{c\, E[n]} = \frac{\operatorname{std}(n)}{E[n]} = \mathrm{CV} \ \Rightarrow\ \log \operatorname{std}(x) = \log \operatorname{mean}(x) + \log \mathrm{CV},$$

with the coefficient of variation CV fixed by the distribution (about 0.52 for Rayleigh, 1 for exponential, $1/\sqrt{L}$ for Gamma), so the slope of log(std) vs. log(mean) over the patches is $k = 1$ whether the pixel is amplitude or intensity. After a log mapping, $x = \alpha \log(c\, n) + \beta = \alpha \log c + \alpha \log n + \beta$. Within a patch c is a constant, so

$$\operatorname{mean}(x) = \alpha \log c + \alpha\, E[\log n] + \beta,$$
$$\operatorname{Var}(x) = E\big[(\alpha \log n - \alpha\, E[\log n])^2\big] = \alpha^2 \operatorname{Var}(\log n) \ \Rightarrow\ \operatorname{std}(x) = \alpha \operatorname{std}(\log n) \quad (\alpha > 0).$$

The mean moves with c but the standard deviation does not, so $\log \operatorname{std}(x) = \text{const}$ over the patches and $k = 0$.

Table 1: Test 1: slope k and intercept b of $\log \operatorname{std}(x) = k \log \operatorname{mean}(x) + b$.

	slope k	intercept b
linear (amplitude or intensity), theory	1	$\log \mathrm{CV}$
log, theory	0	$\log(\alpha \operatorname{std}(\log n))$
HRSID sea (fit)	1.17	-1.17

Amplitude vs. intensity. Following the product specifications, Sentinel-1 and TerraSAR-X/TanDEM-X store amplitude [2, 3], so the HRSID pixel value is taken as amplitude.

Conclusion.

- Linear: confirmed ($k = 1.17$, close to 1 and far from 0).
- Amplitude: according to the product specifications.

1.1 References

[1] S. Wei, X. Zeng, Q. Qu, M. Wang, H. Su, and J. Shi, "HRSID: A high-resolution SAR images dataset for ship detection and instance segmentation," *IEEE Access*, vol. 8, pp. 120234–120254, 2020.

[2] N. Miranda and P. J. Meadows, "Radiometric calibration of S-1 Level-1 products generated by the S-1 IPF," ESA Tech. Note ESA-EOPG-CSCOP-TN-0002, Issue 1.0, May 2015. [Online]. Available: `https://sentiwiki.copernicus.eu/__attachments/a_362015d79d4b31d2ec7bfc41d6edf57727461ab4812ae4fbc7d634df98d3db58/ESA-EOPG-CSCOP-TN-0002%20-%20Sentienl-1%20Radiometric%20Calibration%20of%20Products%202015%20-%201.0.pdf?cb=8a5e65784659665b4ad23a67ec403efe`

[3] Airbus Defence and Space, "Radiometric calibration of TerraSAR-X data: Beta naught and sigma naught coefficient calculation," Tech. Note TSXX-ITD-TN-0049, Issue 3.00, Mar. 2014. [Online]. Available: `https://dep1doc.gfz-potsdam.de/attachments/download/365/r465_9_tsx-x-itd-tn-0049-radiometric_calculations_i3.00.pdf`

2 Ship and Sea Amplitude Distribution

Labels. Besides the bounding box, every HRSID ship has a segmentation polygon that follows the hull (Fig. 1a). The box also covers sea: the polygon fills on average only half of its box (0.34 for the large ship in this example). The polygons split the pixels into two classes (Fig. 1b): ship = inside a polygon, and sea = farther than 10 pixels from every polygon. The 10-pixel guard band between them, which holds the ship edges, is not used.

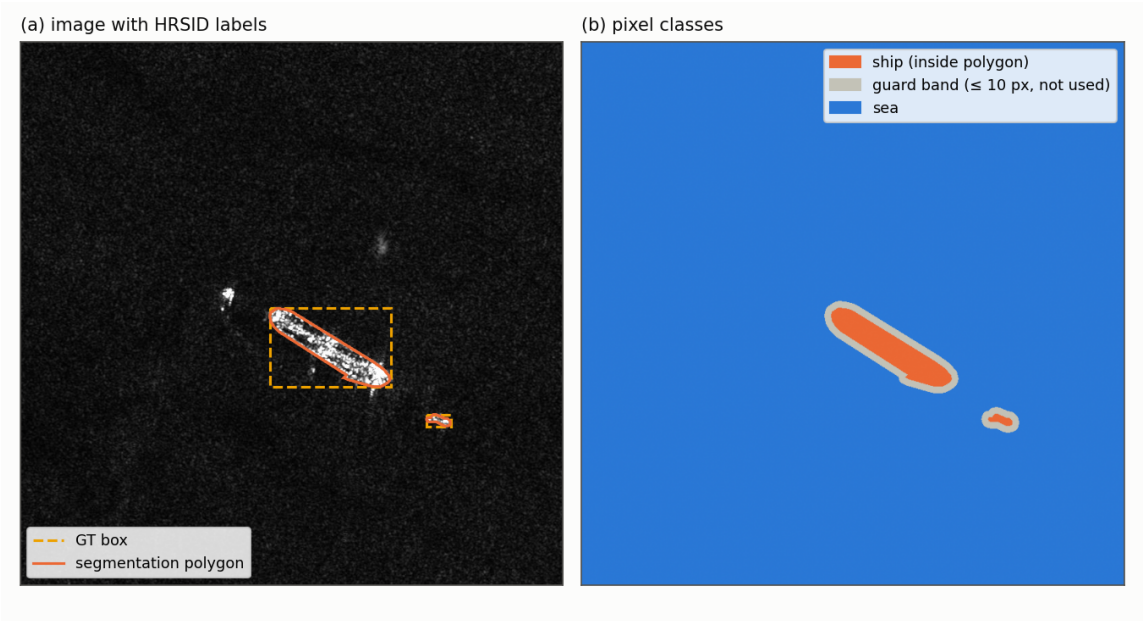


Figure 1: HRSID labels of scene P0126_1200_2000_3000_3800: (a) box and segmentation polygon, (b) pixel classes.

Amplitude histogram. Fig. 2 shows the amplitude distribution of the two classes over all 1593 offshore scenes, each normalized to its own class. The two distributions are clearly different:

- Sea (99.35 % of the pixels): concentrated at low amplitude, median 13, and falls off quickly above about 40.
- Ship (0.20 % of the pixels): spread from about 20 to 255, median 115; 12 % of the ship pixels are clipped at 255.

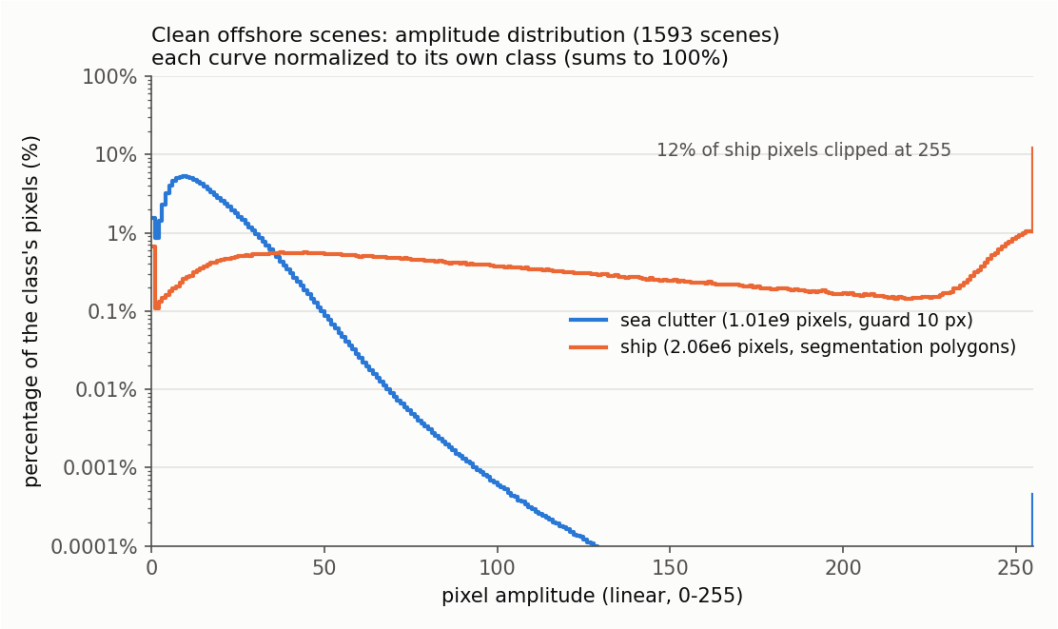


Figure 2: Amplitude distribution of sea and ship pixels (log scale).